

The Equilibrium Fate of Indium in Sphalerite: Large-Scale Replica-Exchange Simulation and the Absence of a Dilute Solid Solution

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Abstract

Whether indium in sphalerite resides in true solid solution or in nanoscale inclusions has resisted empirical resolution because the two states differ at length scales near the limit of detection. We approach it theoretically, asking what configuration the free energy of the cation sublattice actually selects. A preceding study established a configurational Hamiltonian in which the charge-compensation term reduces exactly to a nearest-neighbour pair interaction whose coefficient is fixed by the dielectric constant of ZnS rather than fitted, but could not settle the question: at 4000 sites and 4 at.% solute the solute budget is exhausted by a single domain, so complete condensation was consistent with, but not diagnostic of, a thermodynamic preference.

Here we remove that limitation. Replica exchange with adaptively calibrated ladders reaches states below anything the earlier annealing protocol attained, and we establish the scaling law $\sigma_E \propto x^{0.85} N^{0.51}$ that makes dilute compositions affordable. In a box eight times larger, across a twentyfold range of composition from 0.10 to 2.00 at.%, the solute population condenses completely at every point: the monomer fraction is zero throughout, against a random-solution expectation of 98% monomers at the most dilute point, and the largest cluster differs from the random-solution value by four orders of magnitude.

The result does not depend on convergence at ore-forming temperatures, which the sampler cannot guarantee. Each state point is run from a dispersed and from a pre-condensed configuration, which approach equilibrium from opposite sides and therefore bracket it: at 1800 K, where the acceptance ratio is near 10%, dispersed solutes spontaneously aggregate to 64–95% in one cluster while condensed solutes do not disperse, and the bracket narrows to 100% by 1000 K. Since the ordering preference strengthens as temperature falls, condensation throughout the hydrothermal window follows without further assumption.

Two subsidiary results bear on how trace-element data are read. The pair enrichment factor $R_{\text{Cu-In}}$ rises sixteenfold across the composition scan while the quantity it describes — the copper content of the first shell around indium — falls slightly; the apparent variation lies almost entirely in a normalisation whose ceiling is $1/x_{\text{Cu}}$, an artefact available to be misread in any dataset spanning orders of magnitude in tenor. And the internal structure of the condensed domain is independent of bulk composition to within a surface correction, so the object that forms is the same at every concentration examined, differing only in size.

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The conclusion that the equilibrium state is condensed does not depend on the one class of parameter left unconstrained by first-principles calculation: electrostatics alone gives a sixteenfold enrichment, and removing the chemical term strengthens the claim rather than weakening it, since the resulting energy landscape is smoother. Ideal dilute-mixing entropy, assumed in conventional partition-coefficient models, is therefore not available at ore-forming temperatures. Whether a natural sphalerite grain attains this equilibrium state is a question of kinetics — of interfacial barriers and arrest during cooling — and is the subject of a companion study.

1 Introduction

1.1 An unresolved question of state

Whether indium in sphalerite resides in true solid solution or in nanoscale inclusions of a discrete In-bearing phase is among the longest-standing open questions in trace-element mineralogy, and it is not an academic one. Indium is essentially never mined for its own sake: there are no indium ores in the conventional sense, and the overwhelming majority of primary production is recovered as a by-product of zinc refining (Werner et al., 2015; USGS, 2024). The global indium endowment is therefore controlled not by a distinct ore-forming process but by the trace-element chemistry of sphalerite itself, and the state in which indium sits within that mineral bears directly on whether bulk assay measures a representative concentration, on how the metal partitions during metallurgical recovery, and on whether micro-analytical mapping is a refinement of assay or a measurement of a different quantity altogether.

The question has proved difficult to settle empirically because the two candidate states differ at length scales near the limit of detection. High-resolution LA-ICP-MS imaging and electron microscopy have established that indium is often distributed heterogeneously within single grains, and roquesite (CuInS_2) exsolution from In-rich sphalerite is documented (Moura et al., 2007), but the presence of resolvable inclusions in some samples does not establish the state of indium in the many samples where none are resolved. What has been missing is a theoretical criterion: a statement of which state *should* occur, under which conditions, derived rather than inferred.

1.2 What the preceding work established, and where it stopped

Yu et al. (2026) constructed a configurational Hamiltonian for the sphalerite cation sublattice and showed that the near-1:1 Cu:In molar ratio recorded across deposit types is not a contingent feature of ore-fluid chemistry but a consequence of local electroneutrality on the zinc-blende topology. The central device was an exact topological property: two nearest-neighbour cations share precisely one bridging anion, which reduces the tetrahedral electroneutrality functional without approximation to a nearest-neighbour pair interaction whose coefficient is fixed by the measured dielectric constant of ZnS rather than fitted. The resulting Cu–In pair binding energy, $-2\lambda \approx -0.45$ to -0.73 eV, exceeds $k_B T$ at ore-forming temperatures by an order of magnitude, and the ground state of the charge-compensation term at Cu:In = 1:1 was shown analytically to be exactly the roquesite cation ordering, recovered without any input describing that structure.

That work reported, at 2 at.% substitution, that the solute population condensed into a single connected domain below the ordering crossover. It also stated plainly that this observation could not settle the question of state, and the reason is worth restating because it defines the present paper. At 4000 cation sites and 4 at.% total solute, the 160 solute atoms are exactly sufficient to build one domain of the size the model favours. Once nucleation occurs the entire solute budget is consumed, the order parameters saturate, and the distinction between dispersed pairs, several competing nuclei and a single inclusion is lost. The observation of complete condensation was therefore consistent with, but could not distinguish between, a genuine thermodynamic

preference and an artefact of the box.

Two further limitations compounded this. The Kawasaki acceptance ratio collapses to $\sim 10^{-5}$ below 1300 K, so the sampler could not be trusted to reach equilibrium in the ore-forming window; and all calculations were performed at a single composition lying at the upper end of natural indium contents.

1.3 Hypothesis and approach

We test a specific and falsifiable hypothesis:

Under pure thermodynamic equilibrium at ore-forming temperatures, indium does not form a dilute solid solution in sphalerite at any composition of geological relevance. The equilibrium state is condensed.

The hypothesis is falsifiable in an obvious way. If a dilute solid solution is the equilibrium state at low concentration, then a sufficiently large box at sufficiently low composition must show dispersed solutes, and a composition boundary must exist below which condensation ceases. Finding that boundary would refute the hypothesis as stated and replace it with a solvus; failing to find it down to the lowest composition a box can hold would establish the hypothesis over the range examined.

Three things are required to conduct that test, and the present paper supplies them. First, a sampler that actually equilibrates in the ore-forming window: Section 3 introduces replica exchange with adaptively calibrated ladders, which reaches states below anything the earlier annealing protocol attained. Second, boxes large enough that the solute budget is not the limiting quantity, together with the scaling laws that make such boxes affordable. Third, first-principles values for the parameters that the earlier work left as hypotheses, so that the answer does not rest on assumed interactions.

We emphasise what this paper deliberately does not address. Natural sphalerite grains do sometimes contain several discrete In-rich nanoinclusions rather than one, and equilibrium thermodynamics of the bulk cation sublattice cannot by itself account for a population of separate domains. That observation is a statement about kinetics, interfacial physics and growth history rather than about equilibrium. The resolution we expect, and which a companion paper examines, is kinetic arrest: an ore body cools through a temperature range in which the ordering preference is already saturated (Section 2) while the diffusion that would coarsen several domains into one becomes progressively slower, so that a configuration far from the equilibrium single domain can be frozen in. The equilibrium calculation reported here is a prerequisite for that argument rather than a competitor to it: one cannot identify a state as arrested without knowing what it was arrested short of. The scope here is strictly the equilibrium state: what configuration the free energy selects, at what composition, and with what confidence in the underlying parameters.

2 Theoretical background

This section states the results of Yu et al. (2026) that the present work builds on. Nothing here is new; it is collected so that the argument can be followed without the earlier paper to hand.

2.1 The Hamiltonian and its exact reduction

Sphalerite crystallises in space group $F\bar{4}3m$ with $a_0 = 5.4093$ Å. The cation sublattice is face-centred cubic with twelve first neighbours at $d_{NN} = a_0/\sqrt{2} = 3.825$ Å; sulphur occupies a second

fcc sublattice displaced by $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, so every S is tetrahedrally coordinated by four cations and every cation by four S.

The configurational energy is written as Eq. (4), with the charge-compensation term defined on the anion tetrahedra,

$$E_c = \lambda \sum_a (\Delta Q_a)^2, \quad \Delta Q_a = \sum_{i \in T(a)} \delta q_i, \quad (1)$$

where $T(a)$ is the set of four cations coordinating anion a and δq_i the charge deviation from the host. This is the lattice-discrete form of Pauling’s second rule.

The technical device that makes the model tractable is a topological property of the zincblende structure: two cations are first neighbours if and only if they share exactly one anion tetrahedron, and higher neighbours share none. Expanding the square in Eq. (1) and applying that property gives the *exact* identity of Eq. (5), in which the first term depends only on composition and the second is a strictly nearest-neighbour pair interaction. No approximation or cutoff is involved beyond the initial truncation of the screened Coulomb series from which Eq. (1) is obtained.

2.2 A coefficient that is derived, not fitted

Matching the pair term to the nearest-neighbour term of a screened Coulomb series gives

$$\lambda = \frac{e^2}{8\pi\epsilon_0 \epsilon_r d_{NN}}, \quad (2)$$

which with the static and high-frequency dielectric constants of ZnS brackets $\lambda \in [0.227, 0.368]$ eV. We adopt $\lambda = 0.30$ eV. The resulting binding energy of a first-neighbour Cu–In pair is

$$\Delta E_c^{\text{Cu–In}} = -2\lambda \approx -0.45 \text{ to } -0.73 \text{ eV}, \quad (3)$$

which exceeds $k_B T$ at 300 °C by an order of magnitude. This quantity carries no adjustable parameter, and it is what drives every result reported below.

2.3 The ground state, and why it matters here

At $x_{\text{Cu}} = x_{\text{In}}$ the charge-compensation term has an exact zero-energy ground state: $E_c = 0$ whenever every anion tetrahedron contains two Cu and two In, since then $\Delta Q_a = 0$ identically. That is the adamantine cation-ordering rule of chalcopyrite-structured CuInS_2 , the mineral roquesite, recovered from a purely electrostatic functional with no input describing that structure.

For the present paper the significance is that the model already contains a preferred ordered arrangement of solutes, at zero energy cost, which a dilute solid solution cannot exploit. The question of whether indium disperses or condenses is therefore not a question about whether an ordered state exists — it does — but about whether configurational entropy at realistic dilutions is sufficient to outweigh it. Section 6 answers that quantitatively.

2.4 What was established at 2 at.%, and what was not

Yu et al. (2026) sampled the model by Metropolis Monte Carlo with Kawasaki exchange dynamics at $x_{\text{Cu}} = x_{\text{In}} = 2$ at.% in a 10^3 -cell box. Three results carry over. Electrostatics alone, with all chemical terms switched off, enriches Cu–In first-neighbour bonds sixteenfold over the random solid solution at 300 °C and removes 96% of the local charge imbalance. Elastic misfit, retained at full strength in a control calculation, produces no detectable ordering, the largest

strain-mediated pair interaction being some forty times smaller than $k_B T$. And the ordering crossover lies at a reduced temperature three to seven times above the entire hydrothermal window, so short-range order is saturated wherever sphalerite forms.

What could not be established was the state of indium at realistic dilution, for the reason given in Section 1: the solute budget was exhausted by a single domain. Two sampling limitations compounded this and are addressed in Section 3 and Appendix A.

3 Methods

3.1 The configurational model

We use without modification the four-term configurational Hamiltonian introduced by Yu et al. (2026) for the sphalerite cation sublattice,

$$H = E_c + E_s + E_{sub} + E_{def}, \quad (4)$$

in which E_c penalises departures from local electroneutrality within each sulphur coordination tetrahedron, E_s is the elastic misfit energy in the Eshelby continuum approximation, E_{sub} the substitution energy referenced to the ore fluid, and E_{def} the cation-vacancy term. Cation sites carry a species variable $\sigma_i \in \{\text{Zn}, \text{Cu}, \text{In}, \square\}$ with charge deviations $\delta q_i = \{0, -1, +1, -2\}$ relative to the host.

Two properties of that Hamiltonian are used repeatedly here and are stated for reference. First, because two nearest-neighbour cations in the zinc-blende structure share exactly one bridging anion, the tetrahedral electroneutrality functional reduces *without approximation* to

$$E_c = 4\lambda \sum_i \delta q_i^2 + 2\lambda \sum_{\langle ij \rangle} \delta q_i \delta q_j, \quad (5)$$

so that at fixed composition all configurational physics carried by E_c resides in a strictly nearest-neighbour pair term. Second, the coefficient $\lambda = e^2/8\pi\epsilon_0\epsilon_r d_{NN}$ is fixed by the dielectric constant of ZnS rather than fitted, giving $\lambda \in [0.227, 0.368]$ eV, and we adopt $\lambda = 0.30$ eV throughout.

The chemical term $J_{\alpha\beta}$, which carries short-range interaction beyond the electrostatics already contained in E_c , remains unconstrained by first-principles calculation. Rather than adopt a single value we treat it as an explicit parameter: results are reported at the placeholder values $J_{\text{Cu-In}} = -0.10$ eV and $J_{\text{like}} = +0.05$ eV used by Yu et al. (2026), and every conclusion is classified in Section 6.3 according to whether it survives the removal of J altogether. Section 3.10 sets out how J is treated and what would be required to determine it from first principles.

3.2 Why enhanced sampling is required

Simple Kawasaki exchange dynamics, in which two sites interchange species so that composition is conserved exactly, becomes unusable in the ore-forming window. Yu et al. (2026) report an acceptance ratio falling to $\sim 10^{-5}$ below 1300 K, the direct consequence of a Cu–In pair binding energy of $-2\lambda \approx -0.6$ eV against $k_B T = 0.049$ eV at 300 °C. At that rate essentially every proposed move is rejected and the chain does not decorrelate.

The failure is not merely slow convergence but a convergence test that can return a false positive. Yu et al. (2026) tested equilibration by propagating a quenched and a pre-annealed trajectory at 573 K and observing that their order parameters converged on one another. Replica exchange applied to the same Hamiltonian, box and composition reaches a state 3.1 eV over the box below the lower of those two trajectories, so both were metastable and their mutual agreement carried no information. That correction, its consequences for the values reported in the earlier work, and the general lesson — that in a system with an acceptance ratio of 10^{-5}

the convergence of two trajectories towards each other is not a test of equilibration — are set out in Appendix A.

3.3 Replica exchange

Configurations are sampled by replica exchange (parallel tempering). M replicas are propagated at temperatures $T_1 < \dots < T_M$ and configurations are exchanged between adjacent replicas with probability

$$P_{\text{swap}} = \min[1, \exp((\beta_i - \beta_j)(E_i - E_j))]. \quad (6)$$

Composition is identical across replicas, so no reweighting is required. Within each replica the move set is that of Yu et al. (2026), augmented by a fixed fraction $p_{ss} = 0.5$ of solute–solute exchanges; each component is internally symmetric and the mixing probability is configuration-independent, so the composite proposal remains symmetric.

The solute–solute component deserves comment because it is useless on its own. Measured in a single replica at 573 K, its acceptance is exactly zero: swapping a Cu for an In inside an ordered domain costs of order 4λ , which at $k_B T = 0.049$ eV is unreachable. It becomes useful only inside replica exchange, where the hot replicas accept it freely and pass the decorrelated configurations down the ladder.

Cluster algorithms of the Wolff or Swendsen–Wang type are not applicable here, for two independent reasons. The pair coupling of Eq. (5), $2\lambda \delta q_i \delta q_j$, changes sign with the species pair — negative for heterovalent contacts and positive for homovalent ones — so the model is sign-frustrated and the Fortuin–Kasteleyn mapping does not hold. Separately, cluster flips do not conserve composition, which the canonical ensemble requires. The appropriate generalisation is the geometric cluster algorithm (Dress and Krauth, 1995; Liu and Luijten, 2004), which conserves composition by construction; we have not implemented it, and replica exchange is used instead.

3.4 Ladder construction

A fixed geometric ladder performs badly: with eight replicas over 573–2200 K the minimum pair acceptance is zero. Ladders are therefore built adaptively. A short exchange run measures the acceptance of every adjacent pair, and a new temperature is inserted at the midpoint in β wherever it falls below a target of 0.20; the procedure repeats until every pair clears the target or a replica cap is reached.

The converged ladders are informative in their own right. At $L = 10$ over 573–3000 K the procedure returns 29 replicas concentrated at both ends — nine between 573 and 811 K and eight between 2400 and 3000 K — and sparse in between. The high-temperature cluster coincides with the ordering crossover located by Yu et al. (2026) at 2400–3000 K, where the specific heat peaks and the energy histograms are widest. The low-temperature cluster has a different origin: below the crossover the replicas occupy distinct frozen configurations whose energies differ substantially, so acceptance is poor even though thermal fluctuations are small.

3.5 Convergence diagnostics: round trips, not acceptance

We report the replica round-trip count — the number of times a replica identity travels from the top of the ladder to the bottom and back — as the primary convergence diagnostic, and we recommend it in preference to swap acceptance. The reason is that the two are close to independent. Reducing the interval between exchange attempts from five sweeps to one raises the round-trip count from zero to seven at identical computational cost, while the median swap acceptance moves only from 0.576 to 0.582. A study reporting acceptance alone would have seen nothing.

Figure 1 summarises the three methodological results of this section. Two settings follow from this diagnostic and are used throughout. Exchanges are attempted after every sweep rather than every few sweeps. And the ladder top is placed at 2200 K rather than 3000 K: rebuilding at the lower top requires 19 replicas instead of 29, and the shorter ladder is traversed faster, raising the round-trip count from 7 to 13. The eight replicas above 2400 K bought nothing, because the system is fully disordered well below 3000 K. Across five configurations spanning both settings the internal energy at 573 K agrees to 0.055 meV per site, so the equilibrium value is protocol-independent even where the round-trip statistics are not.

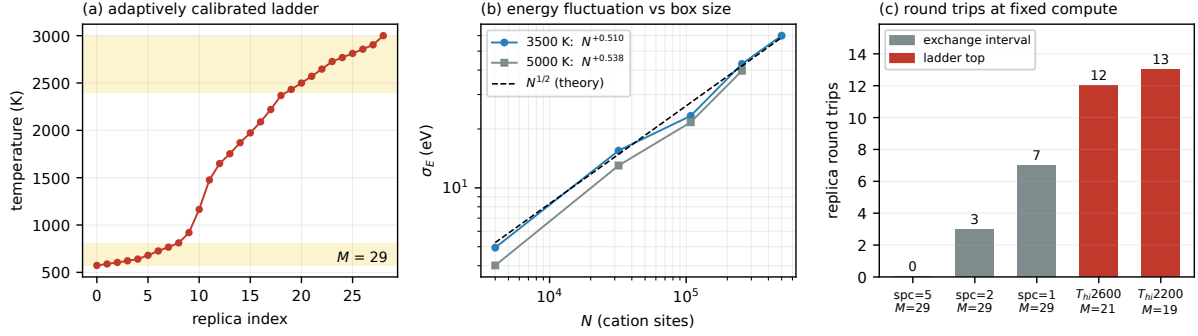


Figure 1: Sampling methodology. (a) A ladder built by adaptive refinement at $L = 10$ over 573–3000 K converges to 29 replicas, concentrated at both ends (shaded) and sparse between: the upper cluster tracks the ordering crossover, the lower one the frozen configurations below it. (b) Equilibrium energy fluctuation against box size, measured in the disordered phase over two decades in N ; the fitted exponents bracket the theoretical $N^{1/2}$ required for $M \propto \sqrt{N}$. (c) Replica round trips at identical computational cost. Attempting exchanges after every sweep rather than every five (grey) takes the count from zero to seven, and lowering the ladder top from 3000 K (grey) to 2200 K (red) raises it to thirteen with ten fewer replicas. Median swap acceptance changes by less than 0.01 across the grey series, which is why round trips rather than acceptance are reported.

3.6 How the ladder length scales

The number of replicas needed over a fixed temperature range is governed by the dimensionless product $(\Delta\beta)\sigma_E$, where σ_E is the width of the energy distribution. Since $\sigma_E^2 = k_B T^2 C_v$ is extensive, $\sigma_E \propto \sqrt{N}$ at fixed composition and $M \propto \sqrt{N}$. We confirm this directly: measured in the disordered phase at 3500 K across five box sizes spanning two decades in N , from 4000 to 5×10^5 cation sites,

$$\sigma_E \sim N^{+0.510}, \quad (7)$$

against the theoretical exponent of +0.500; the measurement at 5000 K gives +0.538.

That result alone is misleading, however, and the distinction matters greatly for the dilute compositions of geological interest. At fixed composition the solute count is proportional to N , so Eq. (7) cannot separate $\sqrt{N}$ from $\sqrt{N_{\text{sol}}}$. Holding the box fixed and varying x instead gives $\sigma_E \sim N_{\text{sol}}^{+0.851}$, which is inconsistent with either simple form. Combining the two measurements,

$$\sigma_E \propto x^{0.85} N^{0.51}, \quad (8)$$

close to the dilute-solution expectation $\sigma_E \propto N_{\text{sol}}/\sqrt{N}$ that follows if the energy fluctuation is carried by the count of solute–solute contacts, which is $\sim N_{\text{sol}}zx/2$ with Poisson-like variance.

Equation (8) reduces the projected cost of a 50^3 campaign at $x_{\text{In}} = 0.1$ at.% — roughly 1000 ppm In, the upper end of natural contents — from about 260 replicas to about 20. It should nonetheless be used for budgeting only, not for setting M . In the very dilute limit it underestimates: at $x = 0.10$ at.% it predicts $M = 8$ while adaptive calibration requires 24, and

the calibrated M is nearly flat at 24–32 for $x \leq 0.5$ at.%, suggesting a floor near 25 independent of composition. The likely reason is that swap acceptance depends not only on σ_E but on the difference in *mean* energy between adjacent replicas, and the energy released on crossing the ordering transition scales with N_{sol} and does not shrink as fast as σ_E does. Equation (8) describes only the denominator. All ladders used in this work are calibrated, not predicted.

3.7 Implementation, scale and verification

The implementation of Yu et al. (2026) is retained, in which both sublattices are held on a single integer grid of spacing $a_0/4$ so that all adjacency is integer table lookup with no floating-point neighbour search. Correctness at the box sizes used here cannot be inherited from that work, which verified the code at up to 6912 sites, and was therefore re-established across $L = 10, 20, 30, 40, 50$, that is up to $N = 5 \times 10^5$ cation sites. The checks are: reciprocity of the adjacency tables; the sharing lemma, tested on random samples of sites at each box size rather than at a single reference site; agreement between the incrementally accumulated energy and a full recomputation *after* a production-length run, which exposes accumulated drift (observed relative deviation $< 10^{-9}$); exact conservation of composition; integrity of the solute index list; and the random-solution limits $R \rightarrow 1$, $R = 1 - \alpha$ and $\langle \Delta Q_a^2 \rangle = 8x$. All pass at every box size.

Resource behaviour sets the practical limits. Memory is not the constraint: the adjacency tables occupy 64 MB at 50^3 and are shared across replicas, with 4.7 MB per replica. Throughput is. Although each trial move is $O(1)$, the partner site is drawn uniformly from all N sites, so every move makes a random access into arrays that leave cache between $N \sim 3 \times 10^4$ and 10^5 ; the measured rate falls as $N^{-0.217}$, reaching 37% of the small-box value at 50^3 (1.37×10^6 trial moves per second, 0.366 s per sweep, single-core). Since replicas evolve independently between exchange attempts and communicate only an energy scalar, parallelisation over replicas rather than optimisation of the serial kernel is the appropriate response: the cache penalty is a factor of 2.7, the replica parallelism a factor of M .

A further limitation of the ladder methodology emerges at scale and is worth recording, because it dictates the workflow of any larger campaign. Calibrating a ladder near the crossover requires equilibrating near the crossover, which is the problem replica exchange exists to solve; the dependency is circular. We resolved it here by measuring the scaling exponent in the disordered phase, where equilibration takes a handful of sweeps, and by keeping all production runs at box sizes where a ladder can be calibrated in advance. At 50^3 the ladder would instead have to be *grown* downwards in temperature during the production run.

3.8 Choice of order parameters

Yu et al. (2026) characterised short-range order through the Warren–Cowley parameter $\alpha_{\text{In-Cu}} = 1 - P_{\text{Cu}}(\text{In})/x_{\text{Cu}}$ and the pair enrichment factor $R_{\text{Cu-In}}$, related by the identity $R_{ab} = 1 - \alpha_{ba}$. Those quantities are unsuitable for the composition scan reported here, and the reason is instructive rather than technical.

Because $R_{\text{Cu-In}} = P_{\text{Cu}}(\text{In})/x_{\text{Cu}}$, its ceiling is $1/x_{\text{Cu}}$, and it approaches that ceiling as the system is diluted. In our scan $R_{\text{Cu-In}}$ appears to explode from 27 to 443 as x falls twentyfold, but the product $R_{\text{Cu-In}} \cdot x_{\text{Cu}}$ — the actual Cu content of the first shell around In — moves only from 0.545 to 0.443. The apparent variation is almost entirely in the normalisation. We therefore report cluster-size distributions and first-shell compositions, and use $\alpha_{\text{In-Cu}}$ and $R_{\text{Cu-In}}$ only for comparison with the earlier work at fixed composition.

Solute clusters are defined by connectivity through the first coordination shell and extracted by a union–find algorithm.

3.9 Simulation protocols

Three families of calculation are reported.

Composition scan. $L = 20$ ($N = 32,000$ cation sites, eight times the volume used by Yu et al. 2026) at $x_{\text{Cu}} = x_{\text{In}} = 0.10, 0.25, 0.50$ and 2.00 at.%, sampled by replica exchange at 573 K with $12,000$ sweeps per replica and ladders of 24 – 69 replicas built by the procedure of Section 3.4. All four points achieved non-zero round trips.

Bracketing runs. To establish whether condensation is an equilibrium property without relying on convergence at 573 K, each state point is run twice, from a random dispersed configuration and from a pre-condensed one, so that equilibrium is bracketed between two approaches from opposite sides. The pre-condensed configuration is a single connected domain grown by breadth-first search over the nearest-neighbour graph and populated by a layered rule giving every interior sulphur tetrahedron two Cu and two In, so that the starting domain is charge-compensated rather than merely compact; connectivity is asserted. Bracketing runs at 1000 , 1400 and 1800 K use plain dynamics, where acceptance is high enough for the chain to be demonstrably mobile; the coarsening runs at 573 K use replica exchange with $36,000$ sweeps per replica.

3.10 The chemical term as an explicit parameter

One class of parameter in Eq. (4) remains unconstrained by first-principles calculation: the short-range chemical term $J_{\alpha\beta}$, which carries interaction beyond the electrostatics already contained in E_c — principally Cu $3d$ –In $5s$ –S $3p$ hybridisation, which has no counterpart in a formal-charge description. Rather than adopt a single value and present results as though it were known, we treat J as an explicit parameter throughout, and classify every conclusion in Section 6.3 according to whether it survives the removal of J altogether.

This is not merely a hedge. The two Hamiltonians differ only in J , so any observable that changes between them is a signature of the chemical term, and three such signatures are identified in this work: the pair enrichment factor $R_{\text{Cu-In}}$, the residual charge imbalance $\langle \Delta Q_a^2 \rangle$, and — less obviously — the ruggedness of the energy landscape, measurable as the replica round-trip count at fixed computational cost (Appendix A.4). The first two respond to E_c in the same direction but to J in opposite directions, which makes them a discriminant that does not require knowing J in advance.

3.10.1 The calculation that would fix it

For completeness we specify the *ab initio* campaign that would determine J , since its design exploits a feature of the problem worth recording. The quantities of interest are *differences* between configurations at the same composition, so the structures can be chosen to be charge neutral and composition matched within each set. No chemical potentials enter, no charged-defect corrections are needed, and the band-gap error that afflicts defect calculations in semiconductors does not propagate. Three sets of $3 \times 3 \times 3$ conventional cells (216 atoms, 108 cation sites) suffice:

Set A one Cu and one In at first, second, third and fourth neighbour separation and at maximum separation. This determines λ and $J_{\text{Cu-In}}$ jointly, and simultaneously tests the strict nearest-neighbour range that the pair reduction of Eq. (5) requires: the binding energy must vanish beyond the first shell.

Set B two Cu and two In in four topologies, which separates J_{like} from $J_{\text{Cu-In}}$. The predicted ground state, in which each Cu is a first neighbour of each In with no homovalent contacts, should lie 4λ below two isolated pairs.

Set C a Zn vacancy with two In in two topologies, giving the configurational part of the competition between the Cu-compensated and vacancy-compensated substitution routes at fixed In content, which is far more robust than the absolute vacancy formation energy.

Parameters would be recovered by least squares over within-set energy differences rather than read off individual structures, using descriptors computed exactly by the lattice code: $\sum_a (\Delta Q_a)^2$ for λ and the first-neighbour bond counts for J . With three free parameters against roughly a dozen independent differences the residual is itself a test of whether the four-term Hamiltonian is adequate. The structures and the fitting harness are part of the released code; the harness was validated by recovering known parameters from synthetic energies carrying 5 meV of noise, and the generated structures reproduce the analytic values of $\sum_a (\Delta Q_a)^2$ predicted by Yu et al. (2026) — 6λ for a first-neighbour Cu–In pair, 8λ for the four-defect ground state, 16λ and 24λ for the vacancy configurations — which is an independent check on the generation procedure before any energies are computed.

We did not run this campaign, and the reason is stated plainly because it bears on what any reader with comparable hardware can reproduce. A 216-atom cell at the cutoffs recommended for these elements requires, by the code’s own estimate, 18.7 GB of memory for a single self-consistent calculation. The workstation used throughout this work has 16 GB installed, of which about 12 GB is available to the calculation. Even were a single job to fit, it would do so with no headroom against the transient peaks that accompany the Hubbard projection and the Davidson subspace, and queueing twelve consecutive jobs on that margin is not a safe use of a machine that has other work to do. Reducing the cutoff far enough to be comfortable degrades the very energy differences the campaign exists to measure, since those differences are small relative to the total energies. The parameterisation is therefore deferred to hardware that can carry it, and Section 4 reports what can be established about J without it.

3.10.2 Iron

Iron would enter the same campaign, in two distinct ways that should not be conflated. Fe^{2+} is isovalent with Zn^{2+} , so $\delta q = 0$ and iron does not enter E_c at all; its influence is through the dielectric response that fixes λ , and through direct $J_{\text{Fe-In}}$ and $J_{\text{Fe-Cu}}$ terms. The first would be obtained by density-functional perturbation theory for $\text{Zn}_{1-x}\text{Fe}_x\text{S}$, delivering a curve $\lambda(x_{\text{Fe}})$ that the Monte Carlo can consume directly. Whether that curve matters is itself a result: if λ varies by less than roughly 15% over the natural iron range, the neglect of iron is retroactively justified.

3.11 Reproducibility

All runs use fixed seeds. Monte Carlo trajectories are not bitwise reproducible across platforms, because the just-in-time compiled random number stream is not guaranteed stable between library versions; the analytic limits of Section 3.7 are, and are asserted at every run. Source code, the numerical results underlying every figure and table, and the scripts that regenerate them are openly available (Section B.3).

4 What is known about the chemical term without computing it

Section 3.10 explains why the *ab initio* determination of $J_{\alpha\beta}$ is deferred. This section establishes what can nonetheless be said about it, and — more importantly for the argument of this paper — which conclusions are independent of it.

4.1 Bounds from the model itself

Three constraints on J follow from calculations reported here and in Yu et al. (2026), without any first-principles input.

The sign of $J_{\text{Cu-In}}$ is negative, and J_{like} positive, if the ordering is to be as observed. Switching the assumed values off entirely reduces $R_{\text{Cu-In}}$ at 573 K and 2 at.% from 25.0 to 16.0, so a covalent stabilisation of the heterovalent pair is required to reach the upper figure. A positive $J_{\text{Cu-In}}$ would reduce the enrichment below the purely electrostatic value, which no reading of the LA-ICP-MS record supports.

The magnitude is bounded above by the local charge imbalance. $R_{\text{Cu-In}}$ and $\langle\Delta Q_a^2\rangle$ respond to E_c in the same direction but to J in opposite directions: switching J on takes $R_{\text{Cu-In}}$ from 16.0 to 25.0 while simultaneously taking $\langle\Delta Q_a^2\rangle$ from 0.0065 to 0.0100, a 54% degradation of local electroneutrality. The mechanism is the geometry of the fcc cation sublattice, in which 24 triangles pass through every site, so a compact solute domain cannot be two-coloured without homovalent edges: the denser Cu-In network that J buys is paid for in homovalent contacts and hence in electrostatic energy. A $J_{\text{Cu-In}}$ much larger in magnitude than the assumed -0.10 eV would drive $\langle\Delta Q_a^2\rangle$ towards the random-solution value and destroy the charge compensation that the Cu-In correlation exists to provide.

A third signature exists in the dynamics. The full model achieves five replica round trips where the electrostatics-only model achieves thirteen at equal computational cost (Appendix A.4). Landscape ruggedness is therefore an observable of J , and unlike the other two it is a property of the sampling rather than of the equilibrium state. We are not aware of this being used as a diagnostic elsewhere, and note it because it is available in any replica-exchange study at no extra cost.

4.2 An experimental route

Because $R_{\text{Cu-In}}$ and $\langle\Delta Q_a^2\rangle$ respond oppositely to J , measurements sensitive to short-range order and to local charge state can constrain it without a full calculation. EXAFS coordination numbers around In would give the first shell composition directly; XANES edge positions and bond-valence sums from high-resolution structural refinement bear on the second. A sample in which the Cu content of the In first shell is high *and* the bond-valence sums are close to ideal would indicate a small $|J|$; a high shell occupancy accompanied by measurable valence-sum deviations would indicate a large one. This is a more direct test than the *ab initio* route and does not depend on the choice of functional or Hubbard parameter.

4.3 What does not depend on J

The central claim of this paper — that the equilibrium state is condensed rather than dilute — is independent of J , and Section 6.3 sets out the argument in full. In brief: the pair binding energy of -2λ that drives condensation is fixed by the dielectric constant of ZnS and contains no

adjustable parameter; electrostatics alone gives a sixteenfold enrichment; and the electrostatics-only model condenses more readily than the full model, not less, because its energy landscape is smoother. Removing J strengthens the central claim rather than weakening it.

5 Equilibrium condensation

5.1 The composition scan

The decisive limitation of Yu et al. (2026) was that its 160 solute atoms in 4000 cation sites were exactly enough to build one domain, so complete condensation carried no information about whether condensation is preferred. The scan reported here removes that limitation in both directions at once: the box is eight times larger, and the composition is varied over a twentyfold range down to 0.10 at.%, at which each solute has some five hundred sites of room. It is worth attaching a physical scale to this: with $a_0 = 5.4093$ Å, an $L = 20$ box is $10.8 \times 10.8 \times 10.8$ nm. Indium declines to disperse even when given ten nanometres in which to do so. Table 1 gives the results.

Table 1: Composition scan at $L = 20$ ($N = 32,000$ cation sites), 573 K, replica exchange with 12,000 sweeps per replica. M is the calibrated ladder length and RT the replica round-trip count. $P(\text{Cu}|\text{In})$ is the copper fraction of the first coordination shell of indium, equal to $R_{\text{Cu-In}} x_{\text{Cu}}$.

$x_{\text{Cu}} = x_{\text{In}}$	N_{sol}	M	RT	$\alpha_{\text{In-Cu}}$	$R_{\text{Cu-In}}$	$P(\text{Cu} \text{In})$	clusters	largest
0.10 %	64	25	2	-440.6	442.7	0.443	1	100 %
0.25 %	160	24	2	-195.7	197.5	0.494	1	100 %
0.50 %	320	32	8	-103.2	103.7	0.518	1	100 %
2.00 %	1280	69	5	-26.5	27.3	0.545	2	85.2 %

The second row is a controlled comparison with the earlier work: the same 160 solute atoms, in eight times the volume, still condense completely. The first row halves the concentration again with the same outcome. Complete condensation is therefore not an artefact of an exhausted solute budget.

The magnitude of the effect is best appreciated against the random-solution expectation rather than against the earlier simulations. At $x = 0.10$ at.% the probability that a given solute has no solute among its twelve first neighbours is $(1 - 2x)^{12} = 0.976$, so a random solid solution at this dilution would consist of 98 % isolated monomers and its largest cluster would contain a few atoms at most, some 3 % of the solute population. The simulated largest cluster contains 100 %. The two differ by four orders of magnitude (Fig. 2b), and the observed monomer fraction is zero at every composition examined.

5.2 An order parameter that misleads, and one that does not

The pair enrichment factor $R_{\text{Cu-In}}$ rises from 27 to 443 across the scan, an apparent sixteenfold strengthening of Cu–In association on dilution. That reading is wrong, and the error is worth setting out because $R_{\text{Cu-In}}$ and the equivalent Warren–Cowley parameter are standard descriptors in this literature.

Since $R_{\text{Cu-In}} = P(\text{Cu}|\text{In})/x_{\text{Cu}}$, its ceiling is $1/x_{\text{Cu}}$, which is 1000 at the most dilute point. The measured value of 443 is already 44 % of the way to a ceiling that has nothing to do with the physics. The numerator — the actual copper content of the first shell around indium — moves in the opposite direction and only slightly, from 0.545 to 0.443 (Fig. 2a). Almost the entire variation in $R_{\text{Cu-In}}$ resides in the normalisation.

Read through the numerator, the result is a stronger one than the enrichment factor suggests: *the internal structure of the condensed domain is essentially independent of the bulk concentra-*

tion. Over a twentyfold dilution the shell composition changes by a fifth, and the residual trend has a clear origin in surface-to-volume ratio rather than in dilution as such, since larger domains have proportionally fewer atoms in contact with the matrix. The same signature appears in the removal of local charge imbalance, which rises monotonically with domain size from 85.0 % to 94.4 %. Extrapolated to the surface-free limit, ideal chalcopyrite-type ordering would give $P(\text{Cu}|\text{In}) = 2/3$; the domains approach that value from below without reaching it, as a finite domain with a surface must.

We therefore report cluster-size distributions and shell compositions throughout, and use $R_{\text{Cu-In}}$ only where comparison with the earlier work at fixed composition requires it. The general point applies beyond this study: any enrichment factor normalised by bulk concentration will appear to strengthen on dilution whether or not anything physical is changing, and in LA-ICP-MS datasets spanning orders of magnitude in trace-element content the same artefact is available to be misread.

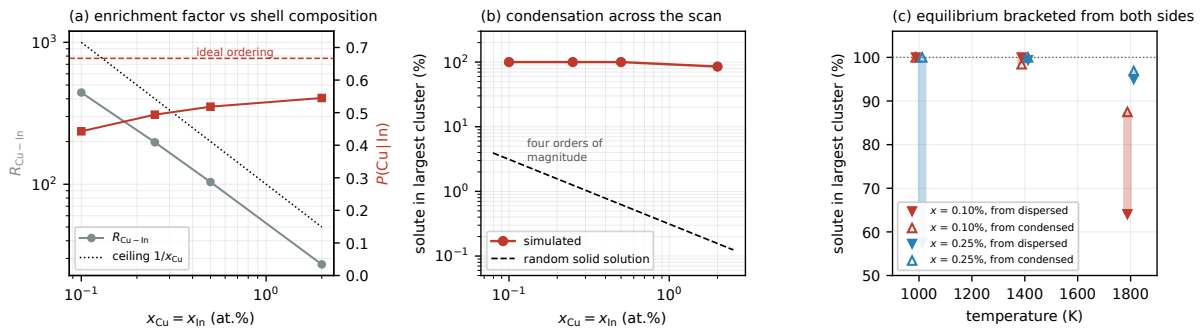


Figure 2: Equilibrium condensation at $L = 20$. (a) The pair enrichment factor $R_{\text{Cu-In}}$ (grey, left axis) rises steeply on dilution, but only because its ceiling $1/x_{\text{Cu}}$ (dotted) rises faster; the copper content of the first shell of indium (red, right axis) is nearly constant and approaches the ideal ordered value of $2/3$ from below as the surface fraction falls. (b) Fraction of solute in the largest cluster, against the expectation for a random solid solution at the same composition; the two differ by four orders of magnitude. (c) The bracketing test. At each temperature the state point is run from a dispersed and from a pre-condensed configuration, which approach equilibrium from opposite sides, so the equilibrium value lies within the shaded interval. Filled triangles start dispersed, open triangles start condensed; points are offset horizontally for clarity.

5.3 The bracketing test

The results of Section 5.1 are subject to the objection that defeated the earlier work. Two of the four state points achieved only two replica round trips, and at 573 K the acceptance ratio within a replica is of order 10^{-5} . A configuration that has not equilibrated may be condensed simply because it was quenched into a condensed basin, and no amount of additional sampling at that temperature demonstrates otherwise if the sampler cannot cross the barriers. Yu et al. (2026) attempted to settle exactly this by propagating two trajectories and observing that they converged on one another, and Appendix A shows that this test returned a false positive: both trajectories were metastable and their agreement carried no information.

We therefore do not rely on convergence at 573 K at all. Instead we exploit a property of Markov chain dynamics that holds whether or not the chain has converged: a chain started on one side of the equilibrium value approaches it monotonically in distribution, so two chains started on *opposite* sides bracket it between them at all times. Each state point is run twice, once from a random dispersed configuration and once from a pre-condensed configuration in which all solutes occupy a single connected, charge-compensated domain (Section 3.9). If the dispersed start aggregates and the condensed start does not disperse, the equilibrium value lies between the two, and this conclusion requires neither run to have converged.

The test is conducted at temperatures where the chain is demonstrably mobile rather than at 573 K, and the bounding argument then transfers downwards: because the ordered state becomes more strongly favoured relative to $k_B T$ as temperature falls, condensation at any temperature below the tested range can be no weaker than within it. Table 2 gives the result.

Table 2: Bracketing test at $L = 20$, 4000 sweeps per run. The entries give the percentage of solute in the largest cluster reached from each starting configuration; the equilibrium value lies between them. Acceptance ratios are quoted to show that the chain is mobile at the upper temperatures.

x	T (K)	acceptance	from dispersed	from condensed	equilibrium within
0.10 %	1800	0.094–0.119	64.1 %	87.5 %	64–88 %
0.10 %	1400	0.007–0.012	100 %	98.4 %	98–100 %
0.10 %	1000	0.0004–0.0006	100 %	100 %	100 %
0.25 %	1800	0.023–0.030	95.0 %	96.9 %	95–97 %
0.25 %	1400	0.002–0.003	99.4 %	100 %	99–100 %
0.25 %	1000	0.0002–0.0003	57.5 % [†]	100 %	—

[†]The single entry at which the two runs fail to bracket usefully, and the one with the lowest acceptance in the table. It is retained rather than omitted; see the text.

The pattern is unambiguous wherever the chain is mobile. At 1800 K, with acceptance near 10 %, solutes starting dispersed *spontaneously aggregate* to between 64 and 95 % in a single cluster, and solutes starting condensed *do not disperse*. The bracket narrows and rises as temperature falls: 64–88 % at 1800 K, 98–100 % at 1400 K, and 100 % at 1000 K (Fig. 2c).

Spontaneous aggregation from a dispersed start is the load-bearing observation. A condensed configuration that merely fails to disperse could be kinetically trapped; a dispersed configuration that assembles itself into a single domain, at an acceptance ratio of 10 % where the chain is manifestly free to do otherwise, cannot be. The two together close the argument from both sides.

One entry does not bracket. At $x = 0.25$ at.% and 1000 K the dispersed start reaches only 57.5 %, against 100 % from the condensed start, and this is the lowest acceptance ratio in the table at 2×10^{-4} . The natural reading is that the dispersed run has not had time to complete aggregation, so its value is a lower bound rather than a bracket endpoint — consistent with the same composition reaching 99.4 % at 1400 K, where the chain moves an order of magnitude faster. We retain the entry rather than omit it, since it marks exactly where the method reaches its limit: bracketing is informative only while both chains can move, and below that the argument must be carried by the transfer downwards in temperature rather than by direct measurement.

5.4 What has been established

Three statements follow, in decreasing order of strength.

The equilibrium state of this Hamiltonian at 0.10 and 0.25 at.% is condensed, established by bracketing at temperatures where the sampler is demonstrably mobile, and transferred downwards to the ore-forming window by the monotonicity of the ordering preference in $1/T$. This does not depend on any low-temperature run having converged.

Complete condensation persists across a twentyfold range of composition in a box eight times larger than that of Yu et al. (2026), with a monomer fraction of zero throughout, against a random-solution expectation of 98 % monomers at the most dilute point. The finite-size objection to the earlier result is therefore answered.

The internal structure of the condensed domain is independent of bulk composition to within the surface correction, so the object that forms is the same object at every concentration examined, differing only in size.

What has *not* been established is where, if anywhere, this behaviour ceases. The compo-

sitions examined here span 0.10 to 2.00 at.%, corresponding to roughly 1000 to 20,000 ppm In, whereas most economically relevant sphalerite carries tens to hundreds of ppm. How far downwards the behaviour persists is taken up in Section 7, where an entropic argument places the composition boundary well below natural tenors and the simulations that would test that estimate are specified.

6 Discussion

6.1 Indium does not form a dilute solid solution

The hypothesis stated in Section 1 was that under thermodynamic equilibrium at ore-forming temperatures indium does not form a dilute solid solution in sphalerite. Over the compositions accessible here, from 0.10 to 2.00 at.%, it is confirmed, and the confirmation does not rest on the convergence of any low-temperature simulation.

It is worth being precise about what a dilute solid solution would look like and how far the observed state is from it. At $x_{\text{In}} = 0.10$ at.% a random distribution of solutes over the cation sublattice leaves 97.6% of them with no solute among their twelve first neighbours, so such a solution would consist almost entirely of isolated substituents, with a largest connected group of a few atoms. The simulated equilibrium state has a monomer fraction of zero and a largest cluster containing every solute in the box. These are not two points on a continuum of ordering; they are different states of matter, separated by four orders of magnitude in the relevant observable.

The physical origin is the pair binding energy of -2λ , which Yu et al. (2026) showed to be fixed by the dielectric constant of ZnS rather than fitted, and which exceeds $k_B T$ at 300 °C by a factor of twelve. A dilute solid solution requires the configurational entropy of dispersal to outweigh that binding energy. At 0.10 at.% the entropy available per solute is of order $k_B \ln(1/x_{\text{tot}})$, where $x_{\text{tot}} = 2x = 0.002$ is the total solute fraction; this is $6.2 k_B$, worth 0.31 eV at 573 K against a binding energy of 0.60 eV per Cu–In pair. Taking x rather than x_{tot} raises the estimate to 0.34 eV and changes nothing. The comparison is not close, and it becomes less close as temperature falls.

6.2 Consequences for trace-element geochemistry

Partition coefficients. Indium partitioning between sphalerite and hydrothermal fluid is conventionally modelled assuming ideal dilute mixing, with configurational entropy $-k_B \sum_{\alpha} x_{\alpha} \ln x_{\alpha}$. If the equilibrium state is condensed, that entropy is not available, and the indium solubility inferred from such models is correspondingly too high. The size of the error is not something the canonical simulations reported here can quantify, because composition is fixed by construction; establishing it requires a grand-canonical treatment in which the solute exchanges with a reservoir at fixed chemical potential. We flag it as a target rather than claim a correction.

Enrichment factors in LA-ICP-MS datasets. The artefact identified in Section 5.2 is not confined to simulation. Any measure of association normalised by bulk concentration — an enrichment factor, a correlation coefficient computed across samples of widely differing tenor, a ratio of observed to expected co-occurrence — has a ceiling set by that concentration, and will appear to strengthen on dilution whether or not the underlying association changes. In our data $R_{\text{Cu-In}}$ rises sixteenfold across a twentyfold dilution while the quantity it is meant to describe, the copper content of the first shell around indium, falls slightly. Datasets spanning orders of magnitude in trace-element content are precisely where this is most available to be misread, and the remedy is to report the un-normalised quantity alongside.

What a condensed state does and does not imply for analysis. A condensed equilibrium state means that indium in a sphalerite grain is not expected to be uniformly distributed at the scale of the cation sublattice. It does not follow that discrete inclusions must be resolvable, nor that bulk assay is unrepresentative: the domains formed here contain 10^2 to 10^3 atoms and are nanometric. What does follow is that micro-analytical mapping and bulk assay measure different quantities in principle, and that the difference between them carries information about the state rather than being analytical scatter.

6.3 Sensitivity to the chemical term

The conclusions above rest on a Hamiltonian containing one class of parameter, $J_{\alpha\beta}$, that remains unconstrained by first-principles calculation. We classify the conclusions accordingly.

Insensitive to J . That the equilibrium state is condensed rather than dilute does not depend on J . With all chemical terms switched off, electrostatics alone enriches Cu–In first-neighbour bonds sixteenfold over the random solid solution at 300 °C and removes 96 % of the local charge imbalance (Yu et al., 2026); the pair binding energy of -2λ that drives condensation is fixed by the dielectric constant and contains no adjustable parameter. Appendix A shows further that the electrostatic model condenses more readily than the full model, not less, since its energy landscape is smoother. Removing J would therefore strengthen the central claim rather than weaken it.

Sensitive to J . The degree of ordering within the domain, the internal Cu:In ratio, and any quantitative composition boundary do depend on J . Switching J on raises $R_{\text{Cu-In}}$ from 16.0 to 25.0 at 2 at.% and simultaneously degrades local electroneutrality, driving the solute-shell Cu:In ratio from its ideal value of 2.00 to 2.67. These are the quantities that an *ab initio* determination of J would fix; Section 4 bounds them in the meantime.

A diagnostic that does not require knowing J in advance. Because $R_{\text{Cu-In}}$ and $\langle \Delta Q_a^2 \rangle$ respond to E_c in the same direction but to J in opposite directions, measurements sensitive to short-range order and to local charge state — EXAFS coordination numbers, XANES edge positions, or bond-valence sums from high-resolution refinement — can constrain the sign and magnitude of the non-electrostatic term without a full *ab initio* calculation. Appendix A.4 identifies a third, independent signature: the ruggedness of the energy landscape, observable as the replica round-trip count at fixed computational cost.

7 Limitations

We set these out in the order in which they bear on the central claim.

The composition range examined stops well above natural tenors. The scan spans 0.10 to 2.00 at.%, roughly 1000 to 20,000 ppm indium, whereas most economically relevant sphalerite carries tens to hundreds of ppm. The entropic argument of Section 6 favours condensation less strongly as dilution proceeds, so a composition boundary must exist in principle, and this work does not locate it. It establishes only that no such boundary lies within the range examined.

The same argument can be pushed further, and the result is worth stating even though it is an order-of-magnitude estimate rather than a measurement. Balancing the dispersal entropy $k_B \ln(1/x_{\text{tot}})$ against the energy a solute gains on joining a domain, which is $2n_b\lambda$ for a solute forming n_b heterovalent bonds, gives

$$x_{\text{tot}}^* \sim \exp\left(-\frac{2n_b\lambda}{k_B T}\right). \quad (9)$$

At 573 K a solute joining the surface of a domain forms one or two heterovalent bonds, and Eq. (9) then places the boundary between roughly 3 ppm and 10^{-5} ppm indium; a solute in a

domain interior forms more, and the estimate falls further still. Even the most conservative of these values lies at or below the crustal abundance of indium, about 0.05 ppm, and all lie far below the tens to hundreds of ppm carried by economically relevant sphalerite.

The implication, if the estimate survives direct calculation, is that no natural sphalerite crystallises above the solvus. Every occurrence of indium in sphalerite would then be supersaturated with respect to condensation, and the question in any particular sample would never be whether the solute is disposed to aggregate but only whether it has had time to. We flag this as a prediction rather than a result: Eq. (9) neglects the entropy of the condensed phase itself and treats n_b as a constant. But it is a prediction this framework makes, and the larger simulations described above would falsify it if it is wrong.

Reaching tens of ppm is a question of hardware rather than of method. A box holding a statistically meaningful number of solutes at 50 ppm requires of order 5×10^5 cation sites, at which the calibrated ladder needs some tens of replicas and a production run costs of order tens of core-days serially. The implementation is verified correct at that box size (Section 3.7) and the scaling laws needed to plan such a campaign are established (Section 3.6), so the obstacle is computational rather than conceptual. All calculations reported here were performed on a single workstation, which we regard as a feature: every result in this paper can be reproduced on ordinary hardware. Locating the composition boundary is left as a well-posed target for a parallel implementation, in which replicas map naturally onto ranks with communication limited to an energy scalar and an occasional configuration exchange.

The chemical term is not determined from first principles. Section 6.3 classifies which conclusions this affects and Section 4 sets out what can be bounded without an *ab initio* calculation, but the quantitative statements about domain-internal ordering should be read as conditional on the assumed values.

The obstacle is hardware and it is worth quantifying, because it defines what a reader with an ordinary workstation can and cannot reproduce. The campaign of Section 3.10 requires 216-atom supercells; at the plane-wave cutoffs recommended for Zn, Cu, In and S the code’s own estimate for a single self-consistent calculation is 18.7 GB, against roughly 12 GB available on a 16 GB workstation. Reducing the cutoff far enough to fit would degrade precisely the energy differences the campaign exists to measure, since those differences are small relative to the total energies. The structures, input files and fitting harness are nonetheless released and validated, so the calculation can be run by anyone with sufficient memory and the parameters dropped into the model without repeating any of the sampling reported here.

Iron is absent. Natural sphalerite commonly carries up to 10 mol% FeS. Fe^{2+} is isovalent with Zn^{2+} and so does not enter the charge-compensation term at all, but it modifies the dielectric response that fixes λ , and direct $J_{\text{Fe-In}}$ and $J_{\text{Fe-Cu}}$ terms need not vanish. Whether this matters is itself a measurable question, framed in Section 3.10 and subject to the same hardware constraint.

The treatment is rigid-lattice and athermal in its elasticity. Atomic relaxation enters only through continuum expressions, which Yu et al. (2026) showed to be numerically negligible for Cu–In because the size mismatch is small; the same would not hold for substituents with large misfit, and the framework should not be transferred to those without re-examination.

Monte Carlo sweeps are not physical time. Nothing in this work constrains the rate at which equilibrium is approached in a cooling ore body. The bracketing construction of Section 5.3 establishes what the equilibrium state *is*; whether a natural sphalerite grain attains it depends on diffusion rates and cooling history, and multi-domain configurations in natural samples are

expected to reflect exactly that gap. That question is outside the scope of this paper and is treated separately.

8 Conclusions

1. Under the configurational Hamiltonian of Yu et al. (2026), the equilibrium state of indium in sphalerite at ore-forming temperatures is condensed, not a dilute solid solution, at every composition examined between 0.10 and 2.00 at.%. The monomer fraction is zero throughout, against a random-solution expectation of 98% at the most dilute point.
2. This is not a finite-size artefact. The same 160 solute atoms that exhausted the box of the preceding study condense completely in eight times the volume, and halving the concentration again, so that each solute has five hundred sites of room, does not change the outcome.
3. The result does not rest on converging any low-temperature simulation. Running each state point from a dispersed and from a pre-condensed configuration brackets the equilibrium value between two approaches from opposite sides, and the transfer to lower temperature follows from the monotonicity of the ordering preference in $1/T$.
4. The pair enrichment factor is a misleading observable in the dilute limit, since its ceiling is $1/x_{\text{Cu}}$. Cluster-size distributions and un-normalised shell compositions should be reported instead, in simulation and in analytical datasets alike.
5. Condensation is driven by electrostatics and does not depend on the unconstrained chemical term. Removing that term strengthens the conclusion, because the resulting energy landscape is smoother and the system condenses more readily.
6. The ideal dilute-mixing entropy assumed in conventional partition-coefficient models is not available if the equilibrium state is condensed, so indium solubilities inferred from such models are too high. Quantifying the correction requires a grand-canonical treatment.
7. What remains open is where this behaviour ceases. Natural tenors run an order of magnitude below the compositions reached here, and locating a composition boundary requires boxes of order 5×10^5 sites. The implementation is verified correct at that scale and the scaling laws are established; the obstacle is computational, not conceptual.

A The equilibration correction to Yu et al. (2026)

A.1 What was claimed

Yu et al. (2026) addressed the possibility that the low-temperature plateau in their order parameters was a consequence of configurational freezing rather than thermodynamic saturation. Their test propagated two trajectories at 573 K, one quenched directly from a random configuration and one pre-annealed through 2000, 1400, 1000 and 800 K. The two converged to $\alpha_{\text{In-Cu}} = -23.17$ and -23.48 respectively, a difference of 1.3 %, with both order parameters stationary over the final 1.2×10^4 sweeps, and the paper concluded that the plateau was thermodynamic in origin.

A.2 Why the test could not support that conclusion

Applying replica exchange to the same Hamiltonian, box and composition, with an adaptively calibrated ladder, gives $H/N = -0.00605$ eV per site from a quenched start within the same

total sweep budget that leaves plain Kawasaki dynamics at -0.00271 eV. That is 0.77 meV per site, or 3.1 eV over the box, below the pre-annealed trajectory, which was the lowest state the earlier work reached. The result is reproduced by independent runs with different seeds, ladder lengths and exchange frequencies; five such configurations agree on the energy to 0.055 meV per site.

Both trajectories of the original test were therefore metastable, and their agreement with one another carried no information about equilibrium. The metastability that the earlier work did report — 3.2 meV per site between the quenched and annealed states — was real, but it was itself an underestimate of the distance to equilibrium.

The general lesson is transferable and is the reason this appendix is included rather than confined to an erratum. *In a system whose acceptance ratio is of order 10^{-5} , the convergence of two trajectories towards each other is not a test of equilibration.* Two chains launched into the same basin will agree with each other whether or not that basin is the equilibrium one. What is required is a sampler capable of crossing the barriers, together with a diagnostic that measures whether it is actually doing so — for replica exchange, the round-trip count of Section 3.5. It is precisely because this failure mode is silent that the present paper rests its central claim on the bracketing construction of Section 5.3, which requires no run to have converged.

A.3 Corrected values, and what they change

Table 3 gives the corrected values at 573 K.

Table 3: Values at 573 K for the full model of Yu et al. (2026), as originally reported from the annealing protocol and as re-measured by replica exchange.

	annealing	replica exchange	shift
H/N (eV per site)	-0.00528	-0.00605	-0.00077
$\alpha_{\text{In-Cu}}$	-23.48	-24.00	2.2 %
$R_{\text{Cu-In}}$	24.48	25.00	2.1 %
$\langle \Delta Q_a^2 \rangle$	0.0120	0.0100	-17 %
charge imbalance removed	92.4 %	93.7 %	—
solute clusters	1 (160 atoms)	1 (160 atoms)	unchanged
In first shell, Cu / In / Zn	0.490 / 0.183 / 0.327	0.500 / 0.188 / 0.312	—

The order parameters shift by about 2%, in the direction of insufficient ordering, as under-convergence requires. No conclusion of the earlier work is altered by the correction.

Two consequences bear on the present paper. First, the solute-shell composition Cu:In moves only from 2.68 to 2.67 against the ideal 2.00, so the departure from ideal ordering identified there is a real feature of the Hamiltonian and not a sampling artefact. Second, and more useful, the correction is confined to the full model: the electrostatics-only calculation, re-measured with a ladder built for that Hamiltonian and a production run achieving thirteen round trips, returns $\alpha_{\text{In-Cu}} = -14.99$, $R_{\text{Cu-In}} = 15.99$ and $\langle \Delta Q_a^2 \rangle = 0.0065$ — the published values, to every digit reported. Four independent ladders with tops at 2000, 2400, 2800 and 3000 K and lengths of 20 to 29 replicas agree on the energy to 0.001 meV per site.

A.4 Where the sampling difficulty comes from

The two Hamiltonians differ only in $J_{\alpha\beta}$, so that contrast localises the difficulty precisely: the sampling problem is caused by the hypothetical covalent term, not by the charge-compensation term. On the pure E_c landscape the system anneals to equilibrium without difficulty; adding J makes it rugged. The same run achieves thirteen replica round trips for the electrostatic model against five for the full model at equal computational cost.

The mechanism is visible in the bond census. At fixed composition the configurational part of E_c is $2\lambda \sum_{\langle ij \rangle} \delta q_i \delta q_j$, so E_c is minimised by maximising the excess of heterovalent over homovalent solute contacts. The electrostatic model does exactly that, forming an open network with 307 Cu–In bonds and *no* homovalent solute contacts whatever. The full model instead has 480 Cu–In bonds together with roughly 90 Cu–Cu and 90 In–In, buying a 56 % denser Cu–In network at the cost of 4.2 eV of electrostatic energy. The homovalent contacts are not something J tolerates but something the denser packing forces: the fcc cation sublattice contains 24 triangles through every site, so a compact solute domain cannot be two-coloured without homovalent edges. The competition between E_c , which favours an open alternating network, and J , which favours dense aggregation, is what makes the full-model landscape rugged.

This has a practical implication, taken up in Section 4. The ruggedness of the energy landscape, observable as the round-trip count at fixed computational cost, is a third signature of J alongside $R_{\text{Cu-In}}$ and $\langle \Delta Q_a^2 \rangle$, and unlike those two it is a property of the dynamics rather than of the equilibrium state.

B Supplementary sampling diagnostics

B.1 Move sets that do not work

Two natural-looking modifications of the sampler were tried and failed, in ways that are informative rather than merely negative.

Boosted solute–solute exchange. In the proposal of Yu et al. (2026) the partner site is drawn uniformly from all N sites, so at 2 at.% substitution only 4% of trial moves exchange two solutes. Since the slow mode identified there was the reordering of a domain interior, and since a Cu $\leftrightarrow$ In swap reorders the interior without breaking any Cu–In bond, raising that fraction seemed an obvious improvement. It is not: measured in a single replica at 573 K, the acceptance of solute–solute exchange is exactly zero. Swapping a Cu for an In inside an ordered domain destroys the local charge-compensation pattern at a cost of order 4λ , which against $k_B T = 0.049$ eV is unreachable. The move becomes useful only inside replica exchange, where hot replicas accept it freely and pass decorrelated configurations down the ladder; we retain it at $p_{ss} = 0.5$ for that reason alone.

Cluster algorithms. A Wolff or Swendsen–Wang construction does not apply here, for two independent reasons. The pair coupling $2\lambda \delta q_i \delta q_j$ changes sign with the species pair — negative for heterovalent contacts, positive for homovalent ones — so the model is sign-frustrated and the Fortuin–Kasteleyn mapping that underpins those algorithms does not hold. Separately, cluster flips do not conserve composition, which the canonical ensemble requires. The appropriate generalisation is the geometric cluster algorithm (Dress and Krauth, 1995; Liu and Luijten, 2004), which conserves composition by construction; we have not implemented it.

B.2 Why round trips and not acceptance

Table 4 records the experiment behind the recommendation of Section 3.5. All five configurations were run at identical computational cost.

Attempting exchanges after every sweep rather than every five takes the round-trip count from zero to seven while the median swap acceptance moves by less than 0.01. A study reporting acceptance alone would have seen nothing, and would have concluded that its ladder was performing well while no replica was traversing it. Lowering the ladder top from 3000 to 2200 K compounds the improvement, requiring ten fewer replicas and reaching thirteen round trips: the

Table 4: Replica round trips at fixed compute, $L = 10$, 573 K.

M	sweeps per exchange	ladder top (K)	round trips	median swap acceptance
29	5	3000	0	0.576
29	2	3000	3	—
29	1	3000	7	0.582
21	1	2600	12	—
19	1	2200	13	—

replicas above 2400 K were buying nothing, because the system is fully disordered well below 3000 K. All five configurations agree on the internal energy at 573 K to 0.055 meV per site, so the equilibrium value is protocol-independent even where the round-trip statistics are not.

B.3 Verification at scale

The implementation was verified at $L = 10, 20, 30, 40$ and 50, that is up to $N = 5 \times 10^5$ cation sites. The checks are: reciprocity of the adjacency tables; the sharing property of Section 2, tested on random samples of sites at each box size rather than at a single reference site; agreement between the incrementally accumulated energy and a full recomputation *after* a production-length run, which exposes accumulated drift (observed relative deviation $< 10^{-9}$); exact conservation of composition; integrity of the solute index list under solute–solute exchange; and the random-solution limits $R \rightarrow 1$, $R = 1 - \alpha$ and $\langle \Delta Q_a^2 \rangle = 8x$. All pass at every box size.

Throughput falls as $N^{-0.217}$, reaching 37% of the small-box rate at 50^3 (1.37×10^6 trial moves per second, 0.366 s per sweep, single core), because the partner site is drawn uniformly from all N sites and the adjacency arrays leave cache above $N \sim 10^5$. Memory is not the constraint: 64 MB of adjacency tables shared across replicas, 4.7 MB per replica.

Data and code availability

Source code, numerical results and figures for this study are openly available at <https://github.com/Ruqing1963/sphalerite-equilibrium-mc>. Code is released under the MIT licence and data under CC BY 4.0; the manuscript remains the copyright of the authors. The repository includes the verification suite that reproduces every analytic limit quoted here, the adaptive ladder construction, the scripts that regenerate each figure, and the structure generation and fitting harness for the parameterisation campaign of Section 3.10, so that the latter can be run by anyone with sufficient memory and the parameters dropped into the model without repeating any of the sampling reported here.

The lattice model itself, together with the analytic results recapitulated in Section 2, is archived separately with Yu et al. (2026) at <https://github.com/Ruqing1963/sphalerite-lattice-mc>.

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References

Dress, C., Krauth, W. (1995) Cluster algorithm for hard spheres and related systems. *Journal of Physics A* 28, L597–L601.

- Liu, J., Luijten, E. (2004) Rejection-free geometric cluster algorithm for complex fluids. *Physical Review Letters* 92, 035504.
- Moura, M.A., Botelho, N.F., de Mendonça, F.C. (2007) The indium-rich sulfides and rare arsenates of the Sn-In-mineralized Mangabeira A-type granite, central Brazil. *The Canadian Mineralogist* 45, 485–496.
- U.S. Geological Survey (2024) *Mineral Commodity Summaries 2024: Indium*. U.S. Geological Survey, Reston, Virginia.
- Werner, T.T., Mudd, G.M., Jowitt, S.M. (2015) Indium: key issues in assessing mineral resources and long-term supply from linchpin polymetallic deposits. *Applied Earth Science* 124, 213–226.
- Yu, H., Chen, R., Lu, H. (2026) A statistical thermodynamic lattice model for Cu–In coupled substitution in sphalerite, version 2.0. *Zenodo*. <https://doi.org/10.5281/zenodo.21880502>